

Practical Work -8

Application of ML Algorithms and Advanced Methods for Image Analysis Task

Due date: 23.04.2026

Using any of the following benchmarked datasets

- Skin Cancer Dataset:
 - <https://www.kaggle.com/datasets/mahdavi1202/skin-cancer>
- The Oxford-IIIT Pet Dataset
 - <https://www.kaggle.com/datasets/tanlikesmath/the-oxfordiiit-pet-dataset>
- Caltech 101:
 - <https://www.kaggle.com/datasets/ceciliala/caltech-101>
- Fashion-MNIST:
 - <https://www.kaggle.com/datasets/zalando-research/fashionmnist>

Task:

1. Data preprocessing:
 - Perform the following EDA
 - Class distribution analysis
 - Image resolution/statistics
 - Data imbalance detection
 - Apply preprocessing:
 - Resizing, normalization
 - Data augmentation
 - Split dataset: Train / Validation / Test (with justification) results.
K-fold split (5-folds)
2. Experimental design:
 - Baseline & Model Development
 - Implement baseline CNN
 - Train pretrained models: ResNet18 / ResNet50; EfficientNet-B0
 - Compare:
 - ❖ Training time
 - ❖ Accuracy / F1-score / AUC
 - Apply:
 - ❖ Transfer learning vs fine-tuning phases.
3. Apply advanced techniques:
 - Data augmentation study (impact analysis)
 - ❖ Traditional augmentation (zoom, flip, rotation, etc.)
 - ❖ Mixed-based augmentation (Mixup, CutMix, etc.)
 - ❖ Generative AI-based augmentation
 - Model architecture comparison (CNN vs ViT)
 - ❖ CNN pretrained models: ResNet, EfficientNet, VGGNet, DenseNet, etc.
 - ❖ Hybrid models

- Explainability AI (Grad-CAM, SHAP)
- Class imbalance handling (focal loss, weighted sampling) overfitting
 - ❖ Focal loss
 - ❖ Sampling techniques (weighted sampling, SMOTE)
- 4. Performance Evaluation:
 - Calculate and report the following performance evaluation metrics: Accuracy, Precision, Recall, F1-score, and ROC-AUC.
 - Analyze the operational trade-offs between Recall and False Positives.

Note: Summarize your findings in a report, include appropriate visualization plots to support your results on how the model handles malicious vs. benign intents, such as confusion matrix, ROC-curve, PR-AUC, t-SNE, etc.